Maximum Likelihood Estimation

Idea: Choose the value in the parameter space that makes the observed data "most likely".

Example:

Suppose that we flip a biased coin which has the probability of getting "Heads" as either 0.2, 0.3, or 0.8.

Suppose that we flip the coin 20 times and see the results:

H, H, T, H, H, H, H, T, H, H, H, H, H, T, H, H, H, H, H , H

Which of 0.2, 0.3, or 0.8 seems "most likely"?

What if we only flip the coin twice?

Model:

For i=1,2, let

$$X_i = \begin{cases} 1 & \text{, if we get "Heads" on the ith flip} \\ 0 & \text{, if we get "Tails" on the ith flip} \end{cases}$$

Let p=P("Heads" on any one flip)

Then

$$X_1, X_2 \overset{iid}{\sim} \text{Bernoulli}(p)$$

where $p \in \{0.2, 0.3, 0.8\}$

Joint pmf:

$$f(x_1, x_2) = P(X_1 = x_1, X_2 = x_2)$$

$$= P(X_1 = x_1) \cdot P(X_2 = x_2) \quad \text{(indep)}$$

$$= p^{x_1}(1-p)^{1-x_1} I_{\{0,1\}}(x_1) \cdot p^{x_2}(1-p)^{1-x_2} I_{\{0,1\}}(x_2)$$

$$I_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

Tabulated values of the joint pmf:

	(x_1, x_2)			
p	(0,0)	(0,1)	(1,0)	(1,1)
0.2	0.64	0.16	0.16	0.04
0.3	0.49	0.21	0.21	0.06
0.8	0.04	0.16	0.16	0.64

		(x_1, x_2)			
		(0,0)	(0,1)	(1,0)	(1,1)
p	0.2	0.64	0.16	0.16	0.04
	0.3	0.49	0.21	0.21	0.06
	0.8	0.04	0.16	0.16	0.64

- When we observe the data to be (0,0)

  i.e. (“Tails”, “Tails”),

  the value of p that gives the highest joint probability (0.64) is 0.2.

		(x_1, x_2)			
		(0,0)	(0,1)	(1,0)	(1,1)
p	0.2	0.64	0.16	0.16	0.04
	0.3	0.49	0.21	0.21	0.06
	0.8	0.04	0.16	0.16	0.64

- When we observe the data to be (0,1) or (1,0)

  i.e. (“Tails”, “Heads”) or (“Heads”, “Tails”),

  the value of p that gives the highest joint probability (0.21) is 0.3.

		(x_1, x_2)			
		(0,0)	(0,1)	(1,0)	(1,1)
p	0.2	0.64	0.16	0.16	0.04
	0.3	0.49	0.21	0.21	0.06
	0.8	0.04	0.16	0.16	0.64

- When we observe the data to be (1,1)
  i.e. (“Heads”, “Heads”),

  the value of p that gives the highest joint probability (0.64) is 0.8.

- When the data is $(x_1, x_2) = (0, 0)$

  $p = 0.2$ is "most likely"

  (gives highest probability of seeing the data)

- When the data is $(x_1, x_2) = (0, 1)$ or $(x_1, x_2) = (1, 0)$

  $p = 0.3$ is "most likely"

  (gives highest probability of seeing the data)

- When the data is $(x_1, x_2) = (1, 1)$

  $p = 0.8$ is "most likely"

  (gives highest probability of seeing the data)

The maximum likelihood estimator for p is:

$$\hat{p} = \begin{cases} 0.2 & \text{, if } (x_1, x_2) = (0,0) \\ 0.3 & \text{, if } (x_1, x_2) = (0,1) \text{ or } (1,0) \\ 0.8 & \text{, if } (x_1, x_2) = (1,1) \end{cases}$$